

# Value-betting engine: what's built, and how it gets to real money

A Betfair-referenced, Pinnacle-confirmed value detector with single-account Stoiximan placement. The detection core is built and tested; the path to live betting is four deliberate phases.

29 tests passing

Phase 1 complete

5 review fixes shipped

3 integration points open

4.9% edge threshold

● Built & tested

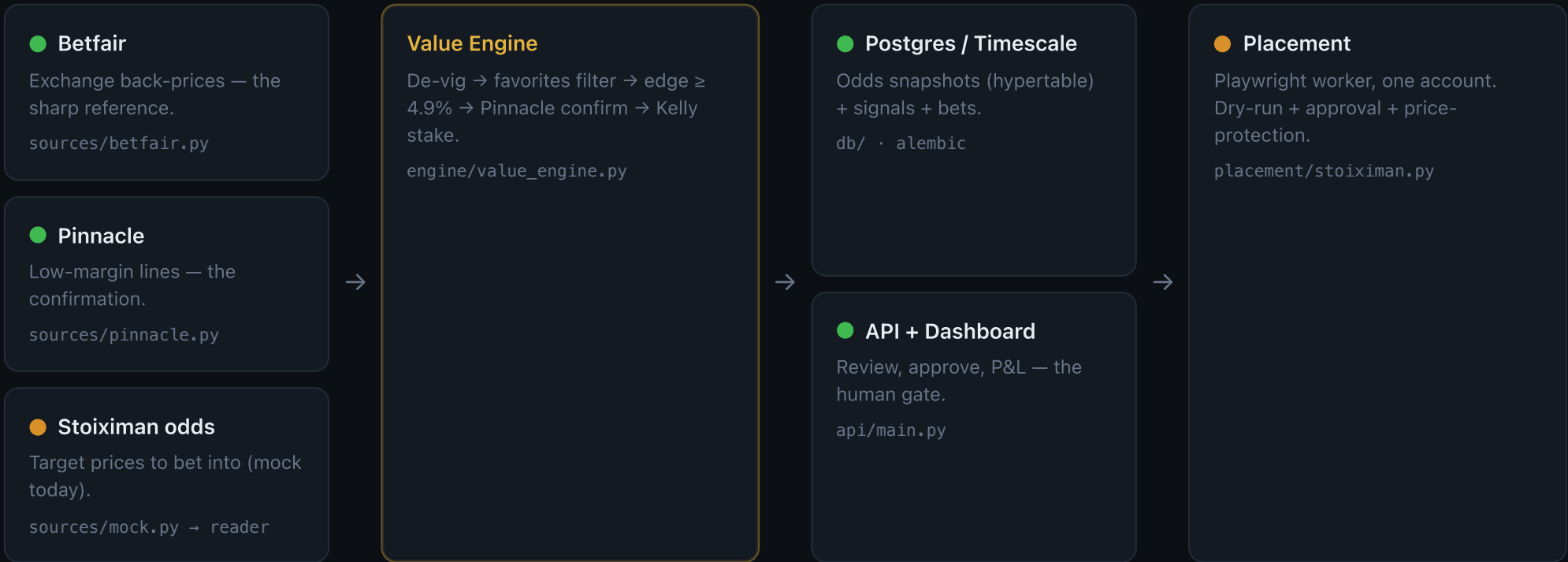
● Scaffolded — needs live site / credentials

○ Planned (roadmap)

## DATA FLOW

### How the system runs, end to end

Every poll, the engine fetches all sources once, finds value, and persists both raw odds and signals. Bets are only ever placed by a human-approved, dry-run-guarded worker.



● **Betfair client**  
`sources/betfair.py`  
Cert login, session keep-alive, MATCH\_ODDS pull.

● **Pinnacle client**  
`sources/pinnacle.py`  
Fixtures + odds join, moneyline extraction.

● **Mock source**  
`sources/mock.py`  
Deterministic feed — runs the whole loop with no credentials.

● **Stoiximan odds reader**  
`planned reader`  
Read-only price scraper. Makes detection real (no public API).

PERSISTENCE & SERVICES

● **DB models + repo**  
`db/`  
Timescale hypertable, signal dedup, deterministic ids.

● **Migrations**  
`alembic / migrations`  
Initial schema + hypertable bootstrap.

● **FastAPI + dashboard**  
`api/`  
Signals, approve/reject, place, settle, P&L.

● **Scheduler + CLI**  
`scheduler/ · cli.py`  
Live vs pre-match polling; init-db / scan / pnl.

PLACEMENT — NEEDS THE LIVE SITE

● **Worker + guards**  
`placement/stoiximan.py`  
Dry-run, approval, price-protection — all wired.

● **Selection navigation**  
`_navigate_to_selection`  
Search event → open market → click selection. Needs live DOM.

● **DOM selectors**  
`SELECTORS map`  
Placeholders to confirm against stoiximan.gr.

DECISION LOGIC

How a signal is decided

A genuine sequence — each gate must pass before the next runs. A price both sharp books disagree with is a trap, not value, so it never reaches sizing.

|    |                                                                             |                                 |
|----|-----------------------------------------------------------------------------|---------------------------------|
| 01 | <b>De-vig Betfair</b><br>Strip the margin → fair probability per selection  | <code>multiplicative</code>     |
| 02 | <b>Favorites only</b><br>Skip anything that isn't a likely winner           | <code>fair_prob ≥ 0.55</code>   |
| 03 | <b>Edge test</b><br>Stoiximan's price must beat the fair price enough       | <code>edge ≥ 4.9%</code>        |
| 04 | <b>Sharp confirmation</b><br>Pinnacle (Shin de-vig) must agree with Betfair | <code> Δp  ≤ 0.03</code>        |
| 05 | <b>Size the bet</b><br>Fractional Kelly, variance-dampened and capped       | <code>½-Kelly, ≤ max</code>     |
| 06 | <b>Human gate</b><br>Surface to dashboard → approve → placement worker      | <code>approval + dry-run</code> |

REVIEW PASS

What we made work better

A correctness sweep after the first build. Five real issues found and fixed — each now covered by a test.

### Single-fetch scanning

**Was:** sources fetched twice per poll — double the API and rate-limit cost.

**Now:** engine fetches once, returns snapshots + signals together.

### Signal de-duplication

**Was:** every poll re-inserted the same opportunity — dozens of dupes/hour.

**Now:** open-signal guard; a live opportunity is stored once.

### Deterministic event ids

**Was:** salted hash( ) gave the same event different ids per run.

**Now:** SHA-256 ids — joins and dedup hold across processes.

### Confirmation enforced

**Was:** with no Pinnacle match, it bet on Betfair alone.

**Now:** no confirmation → no bet (configurable).

### Session reuse

**Was:** re-login to Betfair + new client every single poll.

**Now:** cached sources, session keep-alive — throttle-safe.

## ROADMAP

# The path to live betting

Four phases. We're through the first; the next two make detection real and trustworthy before a cent is staked.

## PHASE 1

● Done

### Detection core & plumbing

- **Odds math, value engine, sources, DB, API, dashboard, scheduler** — all built.
- Validated end-to-end on deterministic mock data; 29 tests green.
- Placement worker with dry-run, approval and price-protection guards.

## PHASE 2 · NEXT

● Make it real

### Real prices in, real slips out

- **Live Stoiximan odds reader** to replace the mock target — the unlock for true edges.
- **Validate DOM selectors** + implement event navigation against the live site.
- Wire **Betfair / Pinnacle / Stoiximan credentials**; run in dry-run against live data.

## PHASE 3

○ Build confidence

### Prove the edge before risking it

- **Backtester** over stored snapshots to calibrate the 4.9% threshold empirically.
- **Closing-line-value (CLV) tracking** — the truest measure of whether the edge is real.
- Calibration & P&L reports to decide go / no-go.

## PHASE 4

○ Go live + operate

### Small stakes, measured, observable

- Flip `PLACEMENT_DRY_RUN=false` — tiny stakes, human approval still on.
- Measure **account durability** — the core question the pilot exists to answer.
- **Redis** scan locks, **Prometheus** metrics, Alembic autogen in CI.

## Bets are placed only when all of these hold

No code path places a real bet without clearing every gate below. This is the system operating one account as a normal user — no multi-accounting, no evasion.

`dry_run` = false

signal **approved** in dashboard

live odds within **slippage**

selectors **validated**

ValueBet pilot — build & status overview. Detection validated on mock data; placement gated behind dry-run until the live Stoiximan integration is confirmed. Automating a bookmaker account may breach its terms; that operating risk is the account holder's.